THE EVEN ICOSAHEDRAL REPRESENTATION OF CONDUCTOR 1951: A CONDITIONAL REDUCTION AND EXPLICIT RESIDUAL BRIDGES

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ABSTRACT. Let ρ be an even icosahedral Artin representation of conductor 1951, the smallest prime conductor of an even icosahedral representation. To our knowledge, Artin's conjecture is not known for any even icosahedral representation of $G_{\mathbb{Q}}$. We record four results. (A) A conditional reduction modelled on Tunnell's proof in the octahedral case: if ρ becomes automorphic over a single quadratic field (for instance an imaginary one), and base change for GL(2) from $\mathbb{Q}$ to the non-normal quintic field L_5 cut out by ρ exists for one specific cuspidal representation, then ρ is modular and $L(\rho, s)$ is entire. (B) The reduction of ρ modulo $\sqrt{5}$ arises, up to a character twist, from an elliptic curve over a number field of degree at most 4 only if that field is totally imaginary and contains $\sqrt{5}$; in particular never over a quadratic field. (C) Explicit elliptic curves realising this twist over $\mathbb{Q}(i, \sqrt{5})$ and over $\mathbb{Q}(\sqrt{5}, \sqrt{-3})$, obtained from Klein's icosahedral resolvent, certified by exact computation, and modular by a theorem of Caraiani and Newton. (D) The reduction of ρ modulo 2 occurs in $S_2(\Gamma_0(1951), \chi)$ through both of its 2-stabilisations, each with multiplicity one, while for parity reasons no holomorphic eigenform is congruent to ρ modulo 4. We compare the base change hypothesis in (A) with the known cubic and quartic cases, including recent work of Wang. The converse-theorem route to it would need automorphic induction from L_5 (for the trivial character, automorphy of the four-dimensional representation $\rho_0 \otimes \rho_0''$ of A_5), so that route is entangled with icosahedral automorphy. Nothing here proves that $L(\rho, s)$ is entire unconditionally.

1. INTRODUCTION

Let $f_5 = x^5 - x^4 - 780x^3 - 1795x^2 + 3106x + 344$, let $L_5 = \mathbb{Q}[x]/(f_5)$, and let M be the Galois closure of L_5, so that $\mathrm{Gal}(M/\mathbb{Q}) \cong A_5$ and M is totally real. Doud and Moore [6] showed that the smallest prime conductor of an even icosahedral Galois representation is 1951. We write $\rho\colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\mathbb{C})$ for the representation of conductor 1951 with projective kernel field M and determinant the character χ of §2, which has order 5 and conductor 1951. Then $\rho = \rho_0 \otimes \chi^3$, where $\rho_0\colon G_{\mathbb{Q}} \to \mathrm{SL}_2(\mathbb{Z}[\zeta_5])$ has image isomorphic to $\mathrm{SL}(2,5)$ and traces in $\mathbb{Z}[\varphi]$, $\varphi = (1+\sqrt{5})/2$. (It is not realisable over $\mathbb{Q}(\sqrt{5})$, since finite subgroups of $\mathrm{SL}_2(\mathbb{R})$ are cyclic.) Complex conjugation acts on ρ by a scalar, so ρ is even.

Artin's conjecture is known for two-dimensional representations with solvable projective image [17, 23], and for odd two-dimensional representations of $G_{\mathbb{Q}}$ as a consequence of Serre's conjecture [13]. For even icosahedral representations of $G_{\mathbb{Q}}$ no case is known [4]. By Booker [2], $L(\rho, s)$ is entire if and only if ρ is modular, i.e. corresponds to a cuspidal automorphic representation of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$; this would be a Maass form of eigenvalue $1/4$, level 1951 and nebentypus χ. There is strong numerical evidence for such a form. Hejhal's method [10] on $\Gamma_0(1951)$, with input only the level, the nebentypus, the parity and the eigenvalue $1/4$, produces a form whose Fourier coefficients agree with the Dirichlet coefficients of $L(\rho, s)$ to within $3 \cdot 10^{-8}$ for $n \le 800$ and $6 \cdot 10^{-7}$ for $n \le 2000$. This is evidence only and is not used below. Booker [3] shows that a Galois-conjugate pair of non-dihedral cusp forms on GL(2) with Hecke eigenvalues in $\mathbb{Z}[\varphi]$ has several properties predicted for icosahedral representations, without producing a Galois representation.

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The purpose of this note is to record what can presently be proved about ρ from the Langlands side, and to state precisely two inputs that would suffice for modularity.

- Theorem A (§3) is a conditional reduction: automorphy of ρ over one quadratic field together with one instance of non-normal quintic base change implies modularity. It is Tunnell's argument [23] moved up one degree, with the central character of each quadratic descent fixed by an adjoint pole argument.
- Theorem B (§4) determines over which fields of degree at most 4 the reduction of ρ modulo $\sqrt{5}$ can come from an elliptic curve.
- Theorem C (§5) gives two explicit modular elliptic curves carrying it.
- Theorem D (§6) describes the reduction modulo 2 inside $S_2(\Gamma_0(1951), \chi)$.

Both hypotheses of Theorem A are open, and nothing in this note proves that $L(\rho, s)$ is entire. Section 7 lists questions on which expert input would be useful. All computations are exact unless labelled numerical; scripts and data are in the supplement.

2. Notation

- $\chi\colon (\mathbb{Z}/1951)^\times \to \mu_5$ is the character with $\chi(3) = \zeta_5^2$, where 3 is a primitive root modulo 1951 (Conrey label 1951.908). It is even, and we also view it as a Galois character.
- $t_p = \operatorname{tr}\rho_0(\mathrm{Frob}_p) \in \{0, \pm 1, \pm 2, \pm\varphi, \pm(\varphi-1)\}$ and $a_p = \operatorname{tr}\rho(\mathrm{Frob}_p) = \chi(p)^3 t_p$ for $p \neq 1951$. The t_p were computed for $p < 3.2 \cdot 10^6$ from Frobenius classes in the field of ρ_0. We used the following inputs.
  - The splitting of p in L_5 and in the field K_{24} of degree 24 fixed by a subgroup of order 5 of $\mathrm{Gal}(\tilde{M}/\mathbb{Q}) \cong \mathrm{SL}(2,5)$, where $\tilde{M}$ is the field of ρ_0. At the primes dividing the index of the defining polynomial (e.g. $2, 7, 71, 137$ for f_5) we used the maximal order.
  - Resolvents separating the two classes of elements of order 5.
  - A local computation at 1951 that distinguishes ρ_0 from its $\sqrt{5}$-conjugate ρ_0''.

  The two choices involved are certified exactly. First, K_{24} is a cyclic quartic extension of K_6 (below) with discriminant $1951^{16} = \mathrm{disc}(K_6)^4$, so it is unramified over K_6 at all finite places; this fixes ρ_0 among its quadratic twists. Second, Theorem D fixes ρ_0 against ρ_0'': modulo 2 the system of $\rho_0'' \otimes \chi^3$ has Serre conductor 1951^2, so it cannot occur in $S_2(\Gamma_0(1951), \chi)$ (cf. Table 1).
- Let g generate tame inertia at 1951. Since ρ has conductor 1951, $\rho(g)$ has eigenvalues $\{1, \chi(g)\}$, so $\rho_0(g)$ has eigenvalues $\{\chi(g)^{-3}, \chi(g)^{-2}\}$ and $\rho_0''(g)$ has eigenvalues $\{\chi(g)^{\pm 1}\}$.
  - Hence $\rho_0'' \otimes \chi^3$ has conductor 1951^2, while $\rho_0'' \otimes \chi$ has conductor 1951 and determinant χ^2.
  - Twisting by a non-trivial quadratic character raises the conductor.
  - So ρ is determined up to isomorphism by M, by $\det\rho = \chi$ and by its conductor.
- $\bar\rho = \rho_0 \bmod \lambda\colon G_\mathbb{Q} \to \mathrm{SL}_2(\mathbb{F}_5)$, where $\lambda = (1-\zeta_5)$ is the prime of $\mathbb{Z}[\zeta_5]$ above 5, so $\lambda^2 = (\sqrt{5})$. Its traces are $t_p \bmod \sqrt{5}$, with $\varphi \equiv 3$, and we call it the reduction modulo $\sqrt{5}$. Reduction is injective on $\mathrm{SL}(2,5)$, since its only non-trivial proper normal subgroup is $\{\pm 1\}$, so $\bar\rho$ is surjective, with projective kernel field M. Let ω_5 be the mod-5 cyclotomic character.
- 2 is inert in $\mathbb{Z}[\zeta_5]$, with residue field $\mathbb{F}_{16} = \mathbb{F}_2(z)$, where $z = \zeta_5 \bmod 2$ and $z^4+z^3+z^2+z+1 = 0$. We write $\bar\rho_2 = \rho \bmod 2\colon G_\mathbb{Q} \to \mathrm{GL}_2(\mathbb{F}_{16})$. Its projective image is $\mathrm{SL}_2(\mathbb{F}_4) \cong A_5$, and $\det\bar\rho_2 = \bar\chi$.
- $F = \mathbb{Q}(i, \sqrt{5}) = \mathbb{Q}[y]/(y^4 + 3y^2 + 1)$ and $K = \mathbb{Q}(\sqrt{5}, \sqrt{-3}) = \mathbb{Q}[a]/(a^4 - a^3 + 2a^2 + a + 1)$. Both have class number 1, and the discriminant of K is 225.
- $\delta = -(10 + 2\sqrt{5}) = -4\sqrt{5}\,\varphi$, so that $\mathbb{Q}(\zeta_5) = \mathbb{Q}(\sqrt{5})(\sqrt{\delta})$; indeed $\sqrt{\delta} = 2(\zeta_5 - \zeta_5^{-1})$.
- K_6 is the sextic resolvent field of f_5, defined by $x^6 - 75x^4 - 70x^3 + 826x^2 - 210x - 625$. It is the fixed field of a dihedral subgroup of order 10 in $\mathrm{Gal}(M/\mathbb{Q})$.

3. A conditional reduction

For a number field E, a cuspidal automorphic representation Π of $\mathrm{GL}_2(\mathbb{A}_E)$ *corresponds to* $\rho|_{G_E}$ if, for almost all places v, Π_v is unramified with Satake parameter the eigenvalues of $\rho(\mathrm{Frob}_v)$.

Hypothesis H1. For a number field E: there is a cuspidal Π on $\mathrm{GL}_2(\mathbb{A}_E)$ corresponding to $\rho|_{G_E}$.

Hypothesis H2. For $k \in \{\mathbb{Q}, \mathbb{Q}(\sqrt{5})\}$ and a given cuspidal π on $\mathrm{GL}_2(\mathbb{A}_k)$: there is an isobaric automorphic representation Π' of $\mathrm{GL}_2(\mathbb{A}_{L_5 k})$ whose local component at almost every place w' of $L_5 k$ is the local base change of π_w, where $w' \mid w$.

Theorem A. (i) *Let K' be a quadratic field. Suppose H1 holds for $E = K'$, and H2 holds for $k = \mathbb{Q}$ and the cuspidal representation π of $\mathrm{GL}_2(\mathbb{A}_\mathbb{Q})$ constructed from it in Lemma 3.3. Then ρ is modular, and $L(\rho, s)$ is entire.*

(ii) *The same conclusion holds if, for some quadratic extension E of $\mathbb{Q}(\sqrt{5})$ (for instance F, K or $\mathbb{Q}(\zeta_5)$), H1 holds for E, and H2 holds for $k = \mathbb{Q}(\sqrt{5})$ and for $k = \mathbb{Q}$, for the representations arising in the proof.*

The proof occupies the rest of this section. Throughout, E/k is a quadratic extension with $k \in \{\mathbb{Q}, \mathbb{Q}(\sqrt{5})\}$, and we write $\chi_k = \chi \circ N_{k/\mathbb{Q}}$ and $\varepsilon = \varepsilon_{E/k}$. Since A_5 has no subgroup of index 2, 3 or 4, the field M has no subfield of degree 2, 3 or 4. So $E \cap M = \mathbb{Q}$, as $[E : \mathbb{Q}] \le 4$, and $\rho|_{G_E}$ still has projective image A_5.

Lemma 3.1 (cyclic towers)**.** *Let $\mathbb{Q} = k_0 \subset k_1 \subset \cdots \subset k_r$ be a tower in which each k_{i+1}/k_i is cyclic. Then $\rho|_{G_{k_r}}$ has projective image A_5. The subgroups of A_5 of index at most 5 are A_5 and the conjugates of A_4.*

Proof. The projective image of $G_{k_{i+1}}$ is normal in that of G_{k_i}, with cyclic quotient. Since A_5 is simple and non-abelian, an image equal to A_5 stays A_5. The second statement is standard. □

Remark 3.2. Consequently, base change and descent along cyclic extensions never reduce ρ to a solvable case. An argument of this kind needs at least one non-normal step to a field over which the projective image becomes solvable. The smallest such field is L_5, where the image is A_4. In the octahedral case the corresponding field is the non-normal cubic field used by Tunnell [23] via [11]. In the icosahedral case no cubic or quartic analogue exists.

Lemma 3.3 (central character of the descent)**.** *Suppose Π_E is cuspidal on $\mathrm{GL}_2(\mathbb{A}_E)$ and corresponds to $\rho|_{G_E}$. Then:*

(a) *$\Pi_E = \mathrm{BC}_{E/k}(\Pi_k)$ for a cuspidal Π_k on $\mathrm{GL}_2(\mathbb{A}_k)$ with central character χ_k;*

(b) *for almost all places w of k, $\Pi_{k,w} \cong \rho_w \otimes \mathrm{unr}(s_w)$ with $s_w \in \{\pm 1\}$, and $s_w = 1$ if w splits in E.*

Proof. By strong multiplicity one, Π_E is $\mathrm{Gal}(E/k)$-invariant, because $\rho|_{G_E}$ is the restriction of a representation of G_k. By Langlands [17] it is therefore the base change of a cuspidal Π_k, unique up to $\otimes\varepsilon$. Its central character ω satisfies $\omega \circ N_{E/k} = \chi_k \circ N_{E/k}$, so $\omega \in \{\chi_k, \chi_k\varepsilon\}$.

Suppose $\omega = \chi_k\varepsilon$. Let w be a place of k inert in E, let $\{\alpha, \beta\}$ be the Satake parameters of $\Pi_{k,w}$, and let $\{a, b\}$ be the eigenvalues of $\rho(\mathrm{Frob}_w)$. Local base change gives $\{\alpha^2, \beta^2\} = \{a^2, b^2\}$, and the central character gives $\alpha\beta = -ab$. Hence $\mathrm{Ad}(\Pi_{k,w})$ has parameters $\{-u, 1, -u^{-1}\}$, where $u = a/b$. At split w we have $\mathrm{Ad}(\Pi_{k,w}) = \mathrm{Ad}\,\rho_w$. Comparing Euler factors gives, for a finite set S of places,

$$L^S(s, \mathrm{Ad}\,\Pi_k) = \zeta_k^S(s)\,\frac{L^S(s, \mathrm{Ad}\,\rho|_{G_k} \otimes \varepsilon)}{L^S(s, \varepsilon)}.$$

Indeed, with $X = q_w^{-s}$, both sides equal $[(1-X)(1+uX)(1+u^{-1}X)]^{-1}$ at inert w.

- The left side is holomorphic and non-zero at $s = 1$. This is because $L^S(s, \Pi_k \times \tilde{\Pi}_k) = \zeta_k^S(s) L^S(s, \operatorname{Ad} \Pi_k)$ has a simple pole there [12]; see also [7].
- On the right, $\operatorname{Ad} \rho|_{G_k} \otimes \varepsilon$ is irreducible and non-trivial, since its image contains A_5. By Brauer induction and Hecke's non-vanishing on $\Re s = 1$, its L-function is holomorphic and non-zero at $s = 1$, as is $L(s, \varepsilon)$. So the right side has a simple pole at $s = 1$.

This contradiction shows $\omega = \chi_k$. The same local computation with $\alpha\beta = ab$ gives $\{\alpha, \beta\} = \pm\{a, b\}$ at inert w, which is (b). □

Lemma 3.4 (the sign is a character)**.** *In the situation of Lemma 3.3, assume H2 for (k, Π_k). Then there is a character μ of $\operatorname{Gal}(E/k)$ such that $\Pi_k \otimes \mu$ corresponds to $\rho|_{G_k}$.*

Proof. Let Π' be the representation of $\mathrm{GL}_2(\mathbb{A}_{L_5 k})$ given by H2.

The tetrahedral form. The projective image of $\rho|_{G_{L_5 k}}$ is A_4, because $L_5 k \cap M = L_5$. By Langlands' theorem for tetrahedral representations over an arbitrary number field [17, §3], there is a cuspidal π_5 on $\mathrm{GL}_2(\mathbb{A}_{L_5 k})$ corresponding to $\rho|_{G_{L_5 k}}$.

Local comparison. By Lemma 3.3(b) and compatibility of local base change with unramified twists, $\Pi'_{w'} \cong \pi_{5,w'} \otimes \operatorname{unr}(s_w^f)$ at almost every place $w' \mid w$, where f is the residue degree of w'/w.

Identifying Π'. Now base change both Π' and π_5 to the quadratic extension EL_5k/L_5k. The two base changes agree at almost all places:

- above places w' inert in EL_5k, the unramified quadratic twist is killed;
- if w' splits in EL_5k, then Frob_w^f fixes E, so either w splits in E (and $s_w = 1$) or f is even.

The representation $\mathrm{BC}(\pi_5)$ is cuspidal, because $\rho|_{G_{EL_5 k}}$ still has projective image A_4. By strong multiplicity one [12], $\mathrm{BC}(\Pi') \cong \mathrm{BC}(\pi_5)$; in particular Π' is cuspidal. By Langlands' description of the fibres of quadratic base change, $\Pi' \cong \pi_5 \otimes (\mu \circ N_{L_5 k/k})$ for a character μ of $\operatorname{Gal}(E/k)$. Here we use that $E \cap L_5 k = k$, as $[L_5 k : k] = 5$ is odd.

Reading off the sign. Let w be a place of k with $\operatorname{tr} \rho(\operatorname{Frob}_w) \neq 0$.

- Then Frob_w has projective order 1, 3 or 5. So every $w' \mid w$ has odd residue degree f, and $\operatorname{tr} \rho(\operatorname{Frob}_{w'}) \neq 0$.
- Comparing the two descriptions of $\Pi'_{w'}$ gives $\pi_{5,w'} \cong \pi_{5,w'} \otimes \operatorname{unr}(s_w^f \mu(w)^f)$.
- This forces $s_w^f = \mu(w)^f$. Otherwise the eigenvalues of $\rho(\operatorname{Frob}_{w'})$ would be of the form $\{a, -a\}$, with trace zero.
- As f is odd, $s_w = \mu(w)$.

If $\operatorname{tr} \rho(\operatorname{Frob}_w) = 0$, then $\rho_w \cong \rho_w \otimes \operatorname{unr}(-1)$ and the sign is irrelevant. Hence $(\Pi_k \otimes \mu)_w \cong \rho_w$ for almost all w. □

Proof of Theorem A. For (i), apply Lemmas 3.3 and 3.4 with $(E, k) = (K', \mathbb{Q})$. For (ii), apply them first with $(E, k) = (F, \mathbb{Q}(\sqrt{5}))$ and then with $(\mathbb{Q}(\sqrt{5}), \mathbb{Q})$. Here H2 over $\mathbb{Q}(\sqrt{5})$ is needed because otherwise $\operatorname{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$ could interchange the two descents of Π_F.

In both cases we obtain a cuspidal π on $\mathrm{GL}_2(\mathbb{A}_\mathbb{Q})$ corresponding to ρ, so $L(\rho, s)$ and $L(\pi, s)$ differ by finitely many Euler factors. The Euler factors of $L(\rho, s)$ have no poles in $\Re s > 0$, the inverse Euler factors of $L(\pi, s)$ are entire, and $L(\pi, s)$ is entire. So $L(\rho, s)$ is holomorphic for $\Re s > 0$, and likewise $L(\rho^\vee, s)$, using $\tilde{\pi}$.

For $\Re s < 1$, write the functional equation as

$$L(\rho, s) = W\, N^{1/2-s}\, \frac{L_\infty(\rho^\vee, 1-s)}{L_\infty(\rho, s)}\, L(\rho^\vee, 1-s), \qquad L_\infty(\rho, s) = \Gamma_\mathbb{R}(s+\nu)^2,$$

with $N = 1951$ and $\nu \in \{0, 1\}$ according as $\rho(c) = \pm I$. Here $1/L_\infty(\rho, s)$ is entire, $L_\infty(\rho^\vee, 1-s)$ has no poles in $\Re s < 1$, and $L(\rho^\vee, 1-s)$ is holomorphic there by the first step. □

Remark 3.5 (status of the hypotheses). (1) *H1.* Over an imaginary quadratic or CM field the parity obstruction disappears, and residual automorphy is available:
- modulo 2 over every imaginary quadratic field, by Theorem D and base change;
- modulo $\sqrt{5}$ over F and K, via the modular elliptic curves of Theorem C. By Theorem B, no elliptic curve over a quadratic field realises it; this does not exclude residual automorphy over a quadratic field by other means.

What is missing is a modularity lifting theorem in the non-cohomological (Artin) weight over a CM field. We know of no such theorem.

(2) *H2.* Non-normal base change for GL(2) is known for cubic extensions [11]. Wang [24] has recently proved base change for GL(2) and GL(3) along non-normal quartic extensions with Galois closure A_4. He uses the converse theorem, automorphic induction to GL(4) via Kim's exterior square lift, and cyclic base change. He also notes that further cases would need base change for GL(4) along non-normal quintic extensions, there with solvable Galois closure (dihedral of order 10, or the Frobenius group of order 20). H2 asks for base change for GL(2) along a non-normal quintic extension whose Galois closure is the non-solvable group A_5.
- When π is known to correspond to a Galois representation, H2 often holds for easier reasons. For π of solvable type it follows from Langlands–Tunnell over L_5k, and for many π attached to elliptic curves over $\mathbb{Q}$ it follows from modularity lifting over the totally real field L_5k. This does not help here: the π in Theorem A is not known to be attached to a Galois representation.
- We know of no case of H2 for π not already known to be of Galois type.

Getz [8] proposes a trace formula approach to non-solvable base change, conditional on conjectural trace identities. Kim's non-normal quintic automorphic induction [14] assumes that the icosahedral representation is automorphic, which is known for odd ones over $\mathbb{Q}$.

(3) *Entanglement.* The converse-theorem route to H2 needs the twists $L(s, \pi \times \mathrm{Ind}_{L_5}^{\mathbb{Q}} \lambda)$ for Hecke characters λ of L_5, and so automorphic induction from L_5 to GL(5). For $\lambda = 1$, $\mathrm{Ind}_{L_5}^{\mathbb{Q}} 1 = 1 \oplus \mathbf{4}$, where $\mathbf{4} \cong \rho_0 \otimes \rho_0''$ is the four-dimensional representation of A_5. For odd icosahedral representations the automorphy of $\mathbf{4}$ follows from that of ρ_0 and ρ_0'' [19], which is how [14] proceeds. For our totally real A_5 field it is open. The case $\lambda = 1$ might be avoided with twists highly ramified at a fixed place, but every other λ needs non-normal quintic automorphic induction, which is equally open. So this route to H2 is entangled with icosahedral automorphy.

4. Where the mod $\sqrt{5}$ reduction can live

Theorem B. *Let K_0 be a number field of degree at most 4. Suppose there are an elliptic curve E'/K_0 and a character $\psi\colon G_{K_0} \to \bar{\mathbb{F}}_5^\times$ with $E'[5] \otimes \bar{\mathbb{F}}_5 \cong \bar\rho|_{G_{K_0}} \otimes \psi$. Then K_0 is totally imaginary and contains $\sqrt{5}$, so it is a CM quartic field. Conversely, if K_0 is a CM quartic field containing $\sqrt{5}$, there is a character $\psi\colon G_{K_0} \to \mathbb{F}_5^\times$ with $\psi^2 = \omega_5|_{G_{K_0}}$.*

Proof. Necessity.

- The Weil pairing gives $\det E'[5] = \omega_5$, and $\det \bar\rho = 1$, so $\psi^2 = \omega_5$.
- At a real place with complex conjugation c, $\omega_5(c) = -1$, while $\psi(c)^2 = \psi(c^2) = 1$. So K_0 is totally imaginary.
- Let $\tilde M$ be the field of ρ_0, with $\mathrm{Gal}(\tilde M/\mathbb{Q}) \cong \mathrm{SL}(2,5)$. Since $\mathrm{SL}(2,5)$ is perfect, $\tilde M \cap \mathbb{Q}(\zeta_5) = \mathbb{Q}$, so $(\bar\rho, \omega_5)$ maps $G_{\mathbb{Q}}$ onto $\mathrm{SL}_2(\mathbb{F}_5) \times \mathbb{F}_5^\times$.
- $\mathrm{SL}_2(\mathbb{F}_5)$ has no proper subgroup of index at most 4. Hence every subgroup of $\mathrm{SL}_2(\mathbb{F}_5) \times \mathbb{F}_5^\times$ of index at most 4 contains $\mathrm{SL}_2(\mathbb{F}_5)$, and $(\bar\rho, \omega_5)$ maps G_{K_0} onto $\mathrm{SL}_2(\mathbb{F}_5) \times \omega_5(G_{K_0})$.

- Suppose $\sqrt{5} \notin K_0$. Then $\omega_5(G_{K_0})$ contains a non-square $c \in \mathbb{F}_5^\times$, so there is $g \in G_{K_0}$ with $\bar\rho(g) = I$ and $\omega_5(g) = c$. Then $\psi(g) \notin \mathbb{F}_5$, and $\operatorname{tr}(\bar\rho \otimes \psi)(g) = 2\psi(g) \notin \mathbb{F}_5$. But the traces of $E'[5]$ lie in $\mathbb{F}_5$, a contradiction.

Sufficiency. If $\zeta_5 \in K_0$, take $\psi = 1$. Otherwise $\omega_5|_{G_{K_0}}$ is the quadratic character of $K_0(\zeta_5) = K_0(\sqrt{\delta})$. A character ψ of order 4 with $\psi^2 = \omega_5$ exists if and only if $K_0(\sqrt{\delta})/K_0$ embeds in a cyclic quartic extension, i.e. if and only if the quaternion algebra $(\delta, -1)$ splits over K_0. Over $\mathbb{Q}(\sqrt{5})$ this algebra behaves as follows:

- It is ramified at both real places, since δ is totally negative.
- It is unramified at finite places not dividing $2\delta = -8\sqrt{5}\varphi$.
- It is split at the place $(\sqrt{5})$, since -1 is a unit there and a square in the residue field $\mathbb{F}_5$.
- By the product formula, it is therefore split at the place above 2 as well.

So it splits over every totally imaginary extension of $\mathbb{Q}(\sqrt{5})$, by the Hasse principle. □

Remark 4.1. For each imaginary quadratic field, Chebotarev provides a single prime certifying the contradiction: p split, $p \equiv \pm 2 \pmod 5$ and $t_p \not\equiv 0 \pmod{\sqrt{5}}$. Here $t_p \equiv 0 \pmod{\sqrt{5}}$ exactly when Frobenius has projective order 2.

- For the 31 fields of discriminant $-d$ with $d \le 100$, the least such prime is at most 53. For example, it is 13 for $\mathbb{Q}(i)$ and 2 for $\mathbb{Q}(\sqrt{-7})$.
- For twelve fields we checked that such primes form a proportion 0.375 ± 0.001 of the split primes $7 \le p < 3.2 \cdot 10^6$, the Chebotarev value 3/8.
- Over F there are none, automatically: degree-one primes of F lie above $p \equiv \pm 1 \pmod 5$.

Once ψ exists, the genus-zero twisting argument of Shepherd-Barron and Taylor [21] produces elliptic curves E' with $E'[5] \cong \bar\rho|_{G_{K_0}} \otimes \psi$, since $\det(\bar\rho \otimes \psi) = \omega_5$. Combined with Theorem B, this shows that among fields of degree at most 4 such curves exist exactly over the CM quartic fields containing $\sqrt{5}$. We do not rely on this: Theorem C gives explicit curves over F and K.

5. Two explicit curves

Klein's resolvent. The trace form on the lattice of trace-zero elements of $\mathcal{O}_{L_5}$ is isometric to the root lattice A_4 with its form multiplied by 1951. For $d \in \{1, 3\}$ there are $u, v \in \mathcal{O}_{L_5}$ with

$$\operatorname{Tr} u = \operatorname{Tr} v = \operatorname{Tr} uv = 0, \qquad \operatorname{Tr} u^2 = d \operatorname{Tr} v^2.$$

For $d = 1$ both norms are 3902:

$$u = \tfrac{1}{19454}\big(-229x^4 - 27x^3 + 180034x^2 + 602971x - 653282\big),$$
$$v = \tfrac{1}{68089}\big(689x^4 - 386x^3 - 535218x^2 - 1566799x + 622376\big).$$

For $d = 3$ the norms are 11706 and 3902:

$$u = \tfrac{1}{136178}\big(3275x^4 - 1143x^3 - 2565972x^2 - 7306095x + 8710406\big),$$
$$v = \tfrac{1}{19454}\big(-145x^4 - 187x^3 + 112806x^2 + 519501x + 192652\big).$$

Then $y = u + \sqrt{-d}\,v$ satisfies $\operatorname{Tr} y = \operatorname{Tr} y^2 = 0$, so it gives a principal quintic over $\mathbb{Q}(\sqrt{-d})$. The square root $\sqrt{-d}$ is Klein's accessory irrationality: since L_5 is totally real, $\operatorname{Tr} y^2 = 0$ cannot be achieved over a real field.

Let $y_0, \dots, y_4$ be the conjugates of y, labelled so that a 5-cycle in $\operatorname{Gal}(M/\mathbb{Q})$ acts by $\nu \mapsto \nu + 1$. The coordinates $c_k = \sum_\nu \zeta_5^{-k\nu} y_\nu$ then satisfy $c_0 = 0$ and $c_1c_4 + c_2c_3 = 0$. On the ruling $z = c_2/c_4$ of this quadric, A_5 acts by icosahedral substitutions [16]. Let f and H be the forms vanishing on the orbits of size 12 and 20, and put $j = 1728\,J$, where $J = \kappa H^3/f^5$ with the constant κ chosen so that $J = 1$ on the orbit of size 30. Thus $j = 0$, 1728, ∞ on the orbits of size 20, 30, 12,

and j is the quotient map $X(5) \cong \mathbb{P}^1 \to X(1)$. The value $j(z)$ is Klein's invariant of y; it lies in $\mathbb{Q}(\sqrt{5}, \sqrt{-d})$.

At 600 digits we obtain the following.

- For $d = 1$: $j_0 = \alpha + \beta\sqrt{5} \in F$ with $\alpha, \beta \in \mathbb{Q}(i)$, with common denominator $29^5 \cdot 41^5 \cdot 1951^2 \cdot 348500149^5 \cdot 696201529^5$.
- For $d = 3$: $j_1 = \alpha' + \beta'\sqrt{5} \in K$ with $\alpha', \beta' \in \mathbb{Q}(\sqrt{-3})$, with common denominator $31^5 \cdot 1951^2 \cdot 22235569^5 \cdot 80579460020178109^5$.

The rational coordinates were recognised with residual below 10^{-590}; the exact values are in the supplement. Theorem C concerns these exact values and does not depend on the recognition step.

Theorem C. *Let E_0/F and E_1/K be elliptic curves with $j(E_0) = j_0$ and $j(E_1) = j_1$, and write (E, K_0) for either pair.*

(i) *The field generated over K_0 by a root of $(X^2 + 250X + 3125)^3 - j(E)X^5$, i.e. the field of definition of a 5-isogeny of E, equals K_0K_6. Consequently the projective representations of $E[5]$ and of $\bar\rho|_{G_{K_0}}$ agree up to conjugation in $\mathrm{PGL}_2(\mathbb{F}_5)$, and $E[5] \otimes \bar{\mathbb{F}}_5 \cong \bar\rho|_{G_{K_0}} \otimes \psi$ for a character ψ.*

(ii) *E has potentially good reduction away from the following places, where $v(j) < 0$:*
- *for E_0: one place above each of 29, 41, 348500149 and 696201529 (with $v = -5$), and the two places above 1951;*
- *for E_1: one place above each of 31, 22235569 and 80579460020178109 (with $v = -5$), and the four places above 1951.*

(iii) *E is modular.*

Proof. (i) The fibre polynomial is irreducible over K_0, and its stem field has degree 24 over $\mathbb{Q}$. A computation with `nfisincl` shows that K_0K_6, also of degree 24, embeds in it, so the two fields are equal. Passing to Galois closures over K_0, the projective 5-torsion field of E is K_0M. Since $\sqrt{5} \in K_0$, both projective representations take values in $\mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$. They are surjective maps $G_{K_0} \to A_5$ with the same kernel, so they differ by an automorphism of A_5, and every such automorphism is conjugation by an element of $\mathrm{PGL}_2(\mathbb{F}_5)$. Two lifts of the same projective representation differ by a character.

(ii) The primes dividing the denominators are those listed. Computing $v_P(j)$ at the places P above them gives the list.

(iii) We use the modularity theorem of Caraiani and Newton for elliptic curves over imaginary CM fields [5, §6.1] at $p = 5$, the result from which their Corollary 6.1.2 is deduced, in the form stated in version 3 of the preprint. There, E over an imaginary CM field K_0 with $\zeta_5 \notin K_0$ is modular if $\bar\rho_{E,5}|_{G_{K_0(\zeta_5)}}$ is absolutely irreducible and $\bar\rho_{E,5}$ is decomposed generic. Decomposed generic means: some prime $\ell \neq 5$ splits completely in K_0, and at every place $v \mid \ell$ the ratio of the eigenvalues of $\bar\rho_{E,5}(\mathrm{Frob}_v)$ is not $\ell^{\pm 1}$. These hypotheses hold here:

- K_0 is an imaginary CM field, Galois over $\mathbb{Q}$, and $\zeta_5 \notin K_0$.
- The field $K_0(\zeta_5)$ is abelian over $\mathbb{Q}$, so it is linearly disjoint from the field $\tilde M$ of ρ_0. Hence $\bar\rho(G_{K_0(\zeta_5)}) = \mathrm{SL}_2(\mathbb{F}_5)$. Twisting by ψ does not change commutators, so the image of $\bar\rho_{E,5}|_{G_{K_0(\zeta_5)}}$ contains $\mathrm{SL}_2(\mathbb{F}_5)$. In particular it is absolutely irreducible.
- Since $\bar\rho_{E,5} \cong \bar\rho|_{G_{K_0}} \otimes \psi$, the eigenvalue ratio at $v \mid \ell$ is that of $\bar\rho(\mathrm{Frob}_\ell)$. For $\ell \equiv 1 \pmod 5$ split in K_0, the condition holds whenever Frob_ℓ has projective order 2 or 3, and such ℓ exist by Chebotarev, as $\tilde M$ and $K_0(\zeta_5)$ are linearly disjoint. Explicit witnesses are also found directly, by computing $a_v(E) \not\equiv \pm(1 + \ell) \pmod 5$ at the four places $v \mid \ell$: $\ell = 89$ for E_0 and $\ell = 19$ for E_1. (Compare Allen–Newton [1, Lemma 2.3].)

□

Remark 5.1 (further checks).
- *Frobenius.* For E_1, the projective order of Frobenius on $E_1[5]$ agrees with that of $\rho(\mathrm{Frob}_p)$ at all 1227 primes of K of good reduction above $p \le 500$, for each of the four conjugates of j_1. Moreover $a_P(E_1)^2 \equiv N(P)\,\bar{t}_p^2 \pmod 5$ at all 383 degree-one primes with $p < 3000$, i.e. $\psi(\mathrm{Frob}_P)^2 = \omega_5(\mathrm{Frob}_P)$. For E_0 the analogous Frobenius check holds at 970 primes of F.
- *Witnesses.* Decomposed-generic witnesses can also be exhibited directly.
  - For $E_0[5]$ there are 30 among the 41 testable split primes below 1200, the smallest being 89. There are 44 split primes in that range: 29 and 41 have a place with $v(j_0) = -5$, and 1069 has a place where $j_0 \equiv 0$.
  - For $E_1[5]$ there are 34 among the 51 split primes below 1500 with good reduction at all four places, the smallest being 19.
- *Genericity rule.* At a split prime ℓ the condition is that the ratio of the Frobenius eigenvalues is not $\ell^{\pm 1}$. For $\ell \equiv 1 \pmod 5$ this excludes Frobenius of projective order 1 and 5; for $\ell \equiv 4 \pmod 5$ it excludes projective order 2.
- *The norm conic.* A point of the conic $x^2 + z^2 = \delta$ over K, such as the one in the supplement, certifies only the existence of ψ. It is not a point of the twisted modular curve.
- *Integral models.* Integral Weierstrass models of E_0 and E_1 are in the supplement.

6. The reduction modulo 2

Let $S = S_2(\Gamma_0(1951), \chi)$, which has dimension 161. Since χ is primitive, every form in S is new. Let $S_{\mathcal{O}}$ be the lattice of forms in S with q-expansion in $\mathcal{O} = W(\mathbb{F}_{16}) = \mathbb{Z}_2[\zeta_5]$, let $\mathbb{T}$ be the $\mathcal{O}$-algebra generated by the Hecke operators acting on it, and let $\bar{S} = S_{\mathcal{O}} \otimes \mathbb{F}_{16}$. The trivial-character space $S_2(\Gamma_0(1951))$, of dimension 162, cannot contain $\bar{\rho}_2$, because $\det \bar{\rho}_2 = \bar{\chi} \neq 1$.

The representation $\bar{\rho}_2$ is unramified at 2, and $\bar{\rho}_2(\mathrm{Frob}_2)$ has the distinct eigenvalues

$$x = z + z^2, \qquad y = 1 + z^3,$$

so that $x + y = z^4 = \bar{a}_2$ and $xy = z^3 = \bar{\chi}(2)$. In weight 2 and level prime to 2, $T_2 \equiv U_2 \pmod 2$ on q-expansions; below, U_2 on $\bar{S}$ means T_2 modulo 2.

Theorem D. (i) *There are normalised forms $G_x, G_y \in \bar{S}$ with $T_\ell G = \bar{a}_\ell G$ for every prime $\ell \nmid 2 \cdot 1951$, and with $U_2 G_x = x G_x$, $U_2 G_y = y G_y$.*

(ii) *The joint eigenspace of T_3, T_5, T_7 in $\bar{S}$ at $(\bar{a}_3, \bar{a}_5, \bar{a}_7)$ is $\mathbb{F}_{16} G_x \oplus \mathbb{F}_{16} G_y$. In particular each of the two $\bar{\rho}_2$-eigenspaces is one-dimensional.*

(iii) *The corresponding generalised eigenspace of T_2, T_3, T_5, T_7 has dimension 9: dimension 3 for T_2-eigenvalue x and 6 for y. Equivalently, the localisations of $\mathbb{T}$ at the two maximal ideals attached to $\bar{\rho}_2$ have ranks 3 and 6 over $\mathcal{O}$. After fixing an embedding $\bar{\mathbb{Q}} \hookrightarrow \bar{\mathbb{Q}}_2$, exactly nine newforms in S are congruent to ρ modulo the maximal ideal of $\bar{\mathbb{Z}}_2$: three with $a_2 \equiv x$ and six with $a_2 \equiv y$.*

(iv) *Let g be a holomorphic Hecke eigenform of any weight and level, and let $\mathcal{O}'$ be the ring of integers of a finite extension of $\mathbb{Q}_2(\zeta_5)$ containing its eigenvalues, with maximal ideal $\mathfrak{m}$. Then $a_p(g) \not\equiv a_p \pmod{2\mathfrak{m}}$ for a set of primes p of positive density.*

Proof. The lattice. The PARI basis of S has $\mathbb{Z}[\zeta_5]$-integral q-expansions, but its reduction modulo 2 has rank only 153. We saturate the lattice by repeatedly dividing by 2 the integral combinations that vanish modulo 2. This is legitimate by Sturm's bound [22], which is $2 \cdot 1952/12 < 326$ in weight 2. Four rounds give rank 161.

(i), first part. The predicted q-expansions are

$$\sum \bar{a}_n q^n, \qquad \bar{a}_{2^j} = x^j \text{ or } y^j,$$

with $\bar{a}_n$ multiplicative and given at odd primes $\ell \neq 1951$ by the weight-2 recursion modulo 2. Both lie in $\bar{S}$ for $n \le 2400$, $n \neq 1951$, i.e. some $G_x, G_y \in \bar{S}$ have these coefficients. Controls

with the $\sqrt{5}$-conjugate traces, or with traces shifted by one prime, do not lie in $\bar{S}$. The forms $T_\ell G - \bar{a}_\ell G$ for $\ell \in \{3,5,7\}$ and $G \in \{G_x, G_y\}$, and $T_2 G_x - x G_x$, $T_2 G_y - y G_y$, lie in $\bar{S}$ and vanish for $n \le 2400/\ell$. Since $2400/7 > 326$, they vanish identically.

(ii) and (iii). The matrices of T_2, T_3, T_5, T_7 on $\bar{S}$ are determined by the coefficients $n \le 326$. They reproduce all coefficients up to $2400/\ell$, and they commute. Kernel computations give:

- dimensions 4, 3, 2 for the joint eigenspaces of T_3, of $\{T_3, T_5\}$ and of $\{T_3, T_5, T_7\}$;
- dimension 1 after adding $T_2 - x$ or $T_2 - y$;
- generalised eigenspaces of dimensions 3 and 6.

Since these kernel dimensions do not change over $\bar{\mathbb{F}}_2$, no maximal ideal of $\mathbb{T}$ other than those of G_x and G_y has the same eigenvalues for $T_2, \dots, T_7$. So the generalised eigenspaces are the full localisations $\bar{S}_\mathfrak{m}$. By the q-expansion duality $S_\mathcal{O} \cong \mathrm{Hom}_\mathcal{O}(\mathbb{T}, \mathcal{O})$, $\dim \bar{S}_\mathfrak{m} = \mathrm{rank}_\mathcal{O} \mathbb{T}_\mathfrak{m}$. Since S consists of newforms, this rank is the number of eigenforms in $S \otimes \bar{\mathbb{Q}}_2$ whose eigenvalues reduce to $\mathfrak{m}$.

(i), all ℓ.

- Every T_ℓ preserves $E' = \mathbb{F}_{16} G_x \oplus \mathbb{F}_{16} G_y$ and commutes with $T_2|_{E'} = \mathrm{diag}(x, y)$. Since $x \ne y$, both G_x and G_y are T_ℓ-eigenforms.
- Their odd coefficients agree everywhere. To see this, let $W = G_x - G_y$ and lift it to $\tilde{W} \in S_\mathcal{O}$.
  - Let $\mathcal{E} = (E_2(\tau) - 1951 E_2(1951\tau))/(1 - 1951) \in M_2(\Gamma_0(1951))$. Its q-expansion is $1 + \frac{4}{325}\sum_{n \ge 1} \big(\sigma(n) - 1951\,\sigma(n/1951)\big) q^n$, so $\mathcal{E} \equiv 1 \pmod 4$.
  - Let τ_0 be the automorphism $\zeta_5 \mapsto \zeta_5^3$ of $\mathcal{O}$, which lifts the inverse Frobenius of $\mathbb{F}_{16}$. Then $\tau_0(T_2 \tilde{W})$ lies in $S_2(\Gamma_0(1951), \chi^3)$.
  - Hence $\Phi = \mathcal{E}\tilde{W} - \tau_0(T_2 \tilde{W})^2$ lies in $S_4(\Gamma_0(1951), \chi)$ and has $\mathcal{O}$-integral coefficients.
  - Modulo 2, Φ has q-expansion $\sum_n w_n q^n - \sum_n w_{2n} q^{2n}$, the odd part of W. It vanishes for all odd $n \le 2400$ with $n \ne 1951$, in particular for $n \le 651$, beyond the weight-4 Sturm bound $4 \cdot 1952/12 < 651$. So $\Phi \equiv 0$.
- By Khare–Wintenberger and Kisin [13, 15] (Serre's conjecture for odd conductor, including $p = 2$), $\bar{\rho}_2$ is the reduction of the Galois representation of a newform h of weight 2 and level $\Gamma_1(1951)$ whose character reduces to $\bar{\chi}$. That character is χ or $\chi\varepsilon_{1951}$, where ε_{1951} is the quadratic character of conductor 1951. As ε_{1951} is odd, $S_2(\Gamma_0(1951), \chi\varepsilon_{1951}) = 0$, so $h \in S$. The reduction of h is an eigenform for all T_ℓ (including T_2) in $\bar{S} \otimes \bar{\mathbb{F}}_2$ with eigenvalues $\bar{a}_\ell$ for $\ell \nmid 2 \cdot 1951$. So it lies in $E' \otimes \bar{\mathbb{F}}_2$ and is a multiple of G_x or of G_y.
- So one of G_x, G_y has eigenvalues $\bar{a}_\ell$ for all $\ell \nmid 2 \cdot 1951$. As their odd coefficients agree, both do.

(iv). Both ρ_g and ρ modulo $2\mathfrak{m}$ have finite image. So the set of $\sigma \in G_\mathbb{Q}$ where their traces differ is a union of conjugacy classes in a finite quotient. It contains complex conjugation:

- $\mathrm{tr}\,\rho_g(c) = 0$, since ρ_g is odd (also when g is an Eisenstein series);
- $\mathrm{tr}\,\rho(c) = \pm 2$, and $2 \notin 2\mathfrak{m}$.

Apply Chebotarev. □

Remark 6.1 (an encoding-free check). This check works in characteristic 0. Let $D_\ell = \det(a_\ell - T_\ell \mid S)$ for odd $\ell \ne 1951$, and $D_2 = \det(T_2^2 - a_2 T_2 + \chi(2) \mid S)$. These are elements of $\mathbb{Z}[\zeta_5]$. We computed them from PARI's Hecke matrices via complex embeddings at 600 digits, and checked them exactly modulo two primes $\equiv 1 \pmod 5$. If $\bar{\rho}_2$ occurs in S, then $2 \mid D_\ell$ for all ℓ. Table 1 records $v_2(D_\ell)$, together with three controls in which $\bar{\rho}_2$ should not occur.

The $\sqrt{5}$-conjugate control differs from the true system exactly at the primes whose Frobenius has order 5 ($\ell = 3, 11, 17, 19$). The shifted control agrees with it only at $\ell = 5$, where the shifted trace coincides with the true one.

TABLE 1. $v_2(D_\ell)$ for $S = S_2(\Gamma_0(1951), \chi)$, and controls: the trivial-character space; the $\sqrt{5}$-conjugate traces $\sigma(t_\ell)\chi(\ell)^3$ (the traces of $\rho_0'' \otimes \chi^3$); and traces shifted to the next prime.

ℓ	2	3	5	7	11	13	17	19	23	29	31	37
S	6	7	12	6	6	6	8	6	6	7	7	6
trivial character	0	0	0	10	0	0	0	0				
$\sqrt{5}$-conjugate	6	0	12	6	0	6	0	0				
shifted	0	0	12	0	0	0	0	0				

Remark 6.2. Both 2-stabilisations occurring in weight 2 is the expected signature of a weight-one Katz form g_1 modulo 2 for $\bar{\rho}_2$, as in Gross's theory of companion forms [9]; one expects $G_x = Ag_1 + yV(g_1)$ and $G_y = Ag_1 + xV(g_1)$, so that $W = (x+y)V(g_1)$. The proof above shows that W has vanishing odd part, i.e. $\theta W = 0$. If Katz's description of the kernel of θ applies at $p = 2$, this would produce g_1. We have not verified this.

Remark 6.3 (the $\sqrt{5}$-conjugate). Let σ be the automorphism $\zeta_5 \mapsto \zeta_5^2$ of $\mathbb{Q}(\zeta_5)$, so that $\sigma(\sqrt{5}) = -\sqrt{5}$, and let $\rho'' = \sigma \circ \rho = \rho_0'' \otimes \chi$. It has conductor 1951, determinant χ^2 and the same projective kernel field M (§2).

- Theorem A holds for ρ'' with the same proof. That proof uses only the projective image, the determinant and the quintic field L_5, which is the same for ρ and ρ''.
- The automorphism σ induces the identity on $\mathbb{Z}[\zeta_5]/\lambda = \mathbb{F}_5$, so $\rho_0'' \bmod \lambda \cong \bar{\rho}$ by Brauer–Nesbitt. Theorems B and C therefore hold verbatim for ρ_0'', with the same curves. A lifting argument starting from E_0 or E_1 would have to distinguish the two lifts ρ_0 and ρ_0'' of $\bar{\rho}$, whose inertial types at 1951 differ.
- Modulo 2, σ induces the Frobenius of $\mathbb{F}_{16}$. The analogue of Theorem D for ρ'' holds in $S_2(\Gamma_0(1951), \chi^2)$, with U_2-eigenvalues $x^2 = 1 + z + z^3$ and $y^2 = 1 + z$ and generalised eigenspaces of dimensions 3 and 6. We also checked this directly in that space.
- We do not know how to deduce the modularity of ρ'' from that of ρ, or conversely.
  - Applying σ to Dirichlet coefficients does not preserve analytic properties. For instance, let ψ be the character modulo 11 with $\psi(2) = \zeta_5$, so that $\sigma \circ \psi = \psi^2$. The first zeros of $L(s, \psi)$ and $L(s, \psi^2)$ on the critical line in the upper half-plane are at heights 5.134 and 4.629.
  - Galois conjugates of Maass forms of eigenvalue 1/4 are not known to be automorphic [20].
  - For odd icosahedral representations over $\mathbb{Q}$ the implication holds, by the rationality of weight-one forms [25]. In the even case Booker [3] and Ramakrishnan [20] take the existence of the conjugate form as a hypothesis.
  - Since $\rho \otimes \rho'' \cong \mathbf{4} \otimes \chi^{-1}$, the route to $\mathbf{4}$ through [19] in Remark 3.5(3) needs the modularity of both ρ and ρ''.

7. QUESTIONS

1. Is Theorem A, in either form, recorded in the literature? Is there an argument for the sign in Lemma 3.4 that avoids non-normal base change?
2. Is any case of base change for GL(2) along a non-normal quintic extension with Galois closure A_5 known for π not already known to be attached to a Galois representation? Is it known conditionally only on automorphy of the four-dimensional representation of A_5, or of automorphic induction from L_5?

3. Is there a route to H1 over an imaginary quadratic field that starts from the mod-2 residual representation? Lifting theorems over CM fields are usually stated for $p > 2$, and Theorem B rules out a mod-5 elliptic starting point over quadratic fields.
4. Have curves such as those in Theorem C, built from Klein's accessory irrationality over CM quartic fields, been recorded for other even icosahedral representations?

Data and code

The supplement contains the scripts and outputs behind every computational statement, with a README that maps each statement to its script and output:

- the trace tables and their generator, and the check that K_{24}/K_6 is unramified at finite places;
- the exact j_0 and j_1;
- the Klein resolvent at 600 digits and the field-equality certificates;
- the Frobenius and witness checks;
- the saturation and Hecke-module computations modulo 2;
- the characteristic-0 determinant check;
- the Hilbert-symbol and certifying-prime computations;
- the Hejhal computation cited in the introduction (numerical).

All exact computations use PARI/GP [18].

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